Data Models, Coordinates, and Geometries

HES 505 Fall 2025: Session 7

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Today's Plan

1. Ways to view the world
2. What makes data (geo)spatial?
3. Geometries, support, and spatial messiness

How do you view the world?

...As a Series of Objects?

- The world is a series of *entities* located in space.
- Usually distinguishable, discrete, and bounded
- Some spaces can hold multiple entities, others are empty
- Objects are digital representations of entities

…As a Continuous Field

Operationalizing views of space

What is a data model?

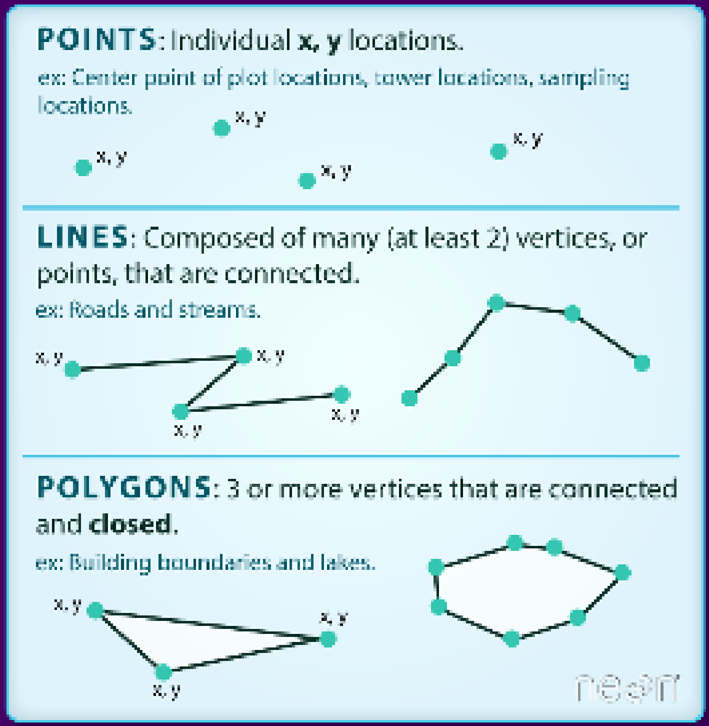
- **Data:** a collection of discrete values that describe phenomena
- Your brain stores millions of pieces of data
- Computers are not your brain
  - Need to organize data systematically
  - Be able to display and access efficiently
  - Need to be able to store and access repeatedly
- Data models solve this problem

2 Types of Spatial Data Models

- **Raster:** grid-cell tessellation of an area. Each raster describes the value of a single phenomenon. More next week…
- **Vector:** (many) attributes associated with locations defined by coordinates

The Vector Data Model

- **Vertices** (i.e., discrete x-y locations) define the shape of the vector
- The organization of those vertices define the *shape* of the vector
- General types: points, lines, polygons



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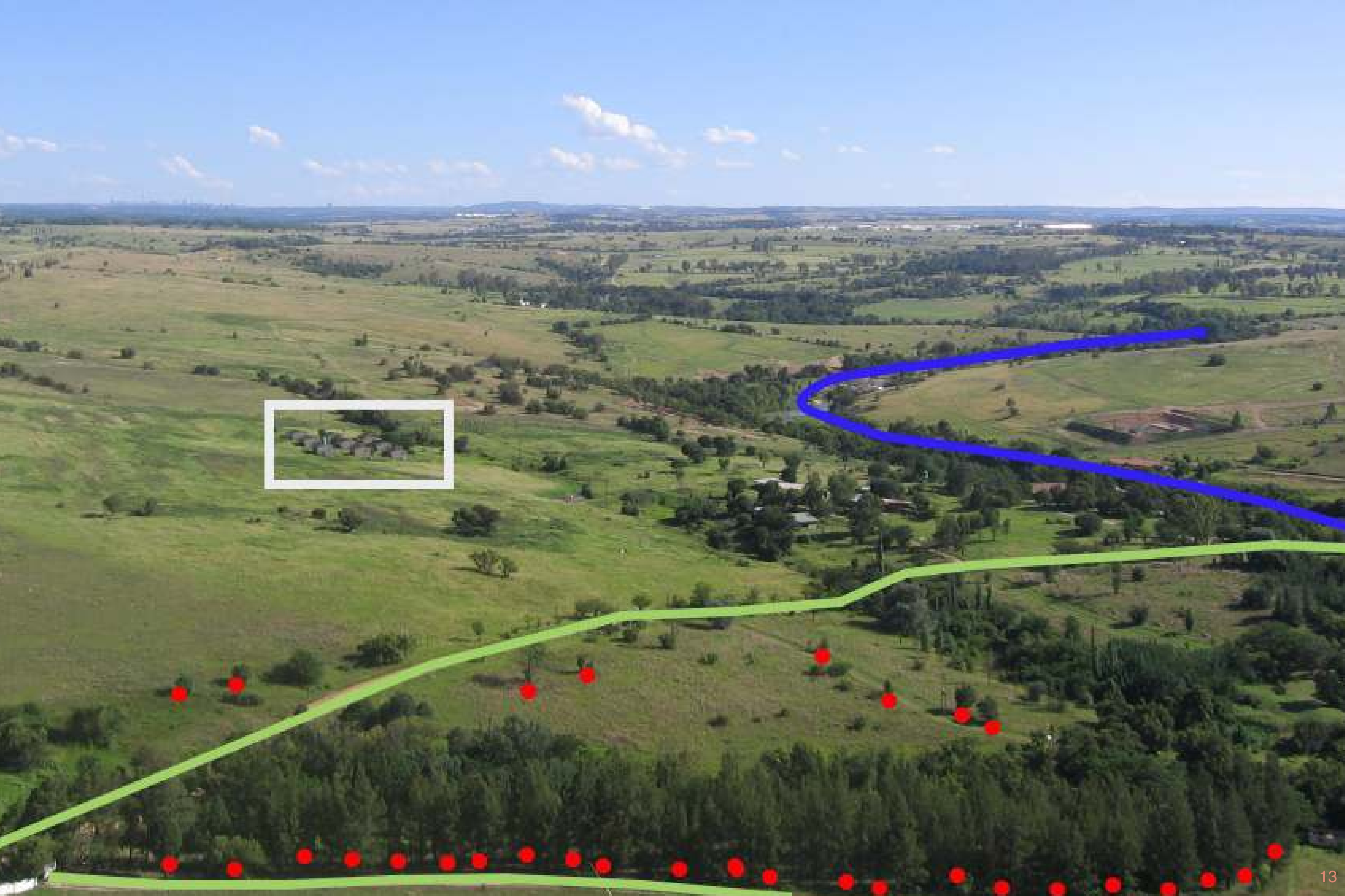
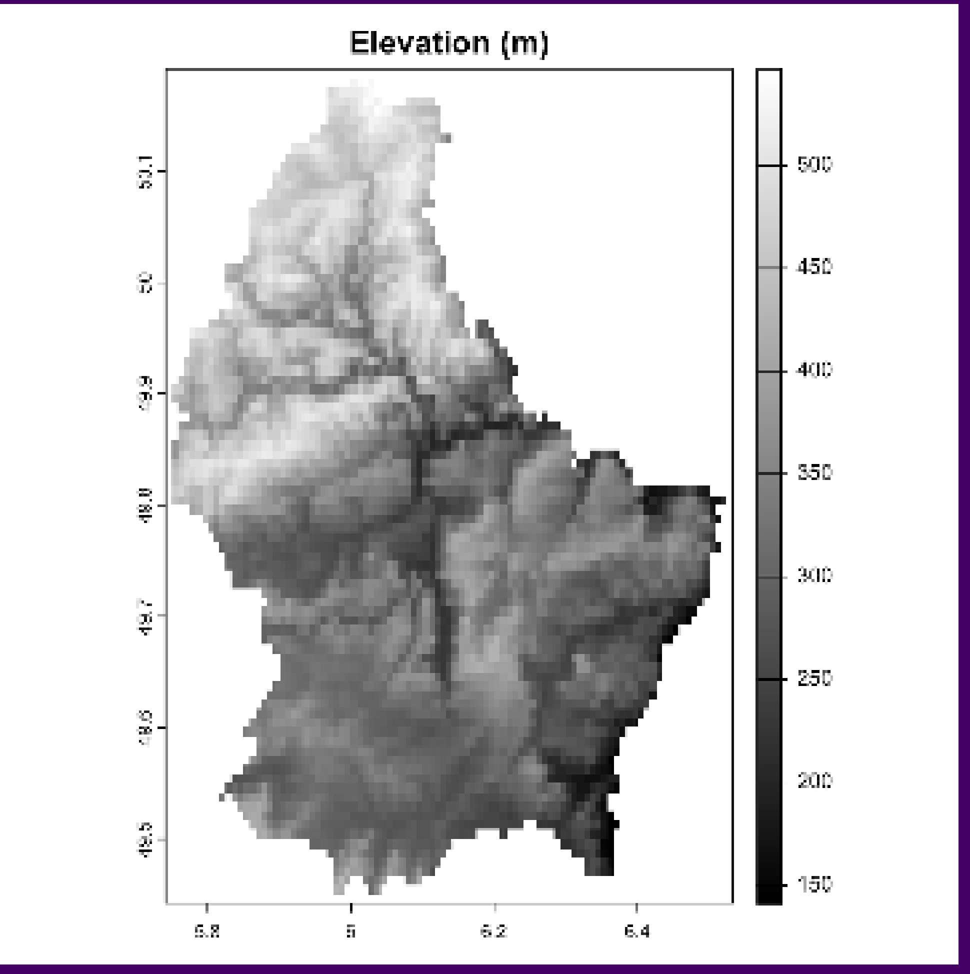


Vectors in Action

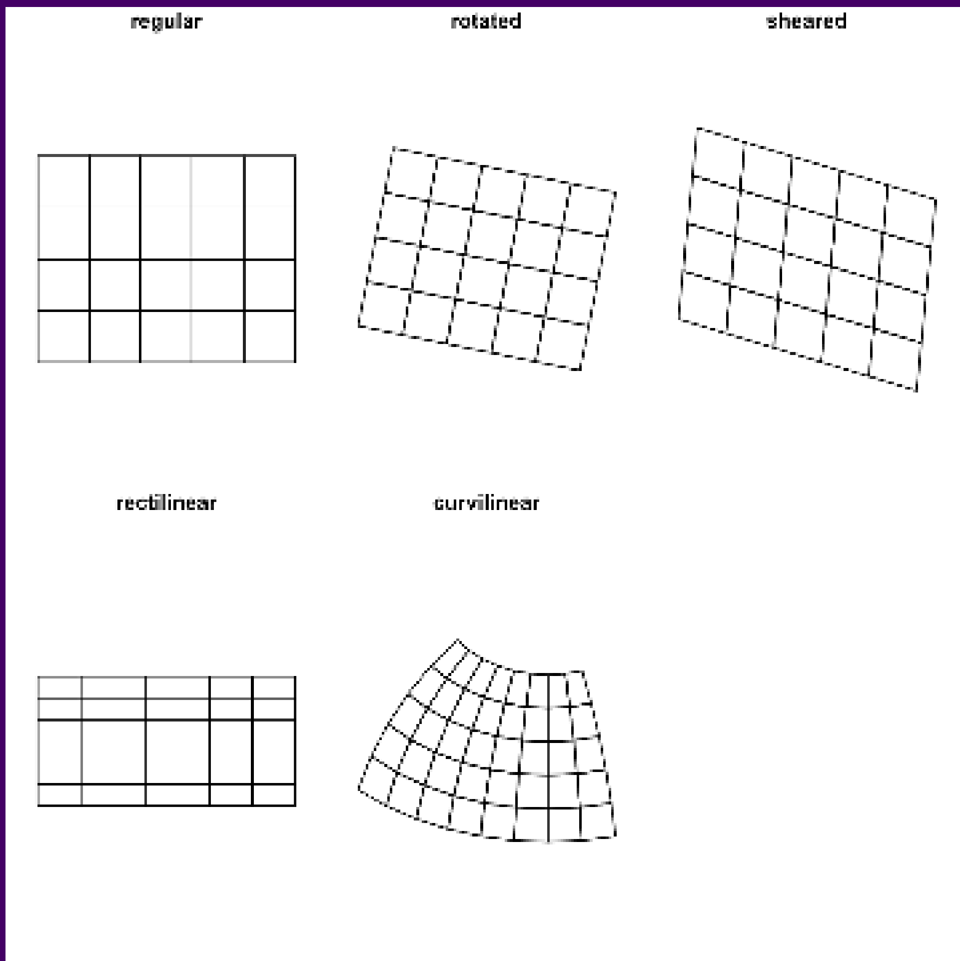
- Useful for locations with discrete, well-defined boundaries
- Very precise (not necessarily accurate)

The Raster Data Model

- **Raster data** represent spatially continuous phenomena (NA is possible)
- Depict the alignment of data on a regular lattice (often a square)
- Geometry is implicit; the spatial extent and number of rows and columns define the cell size



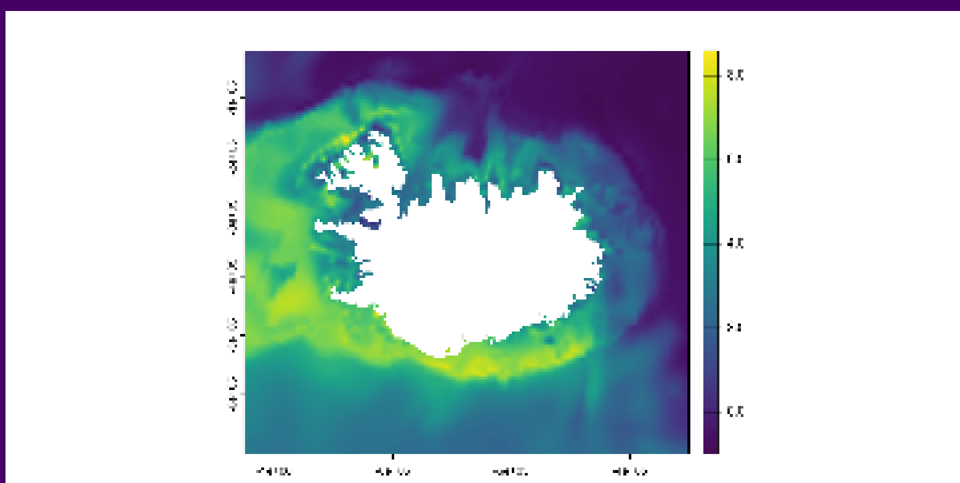
Types of Raster Data

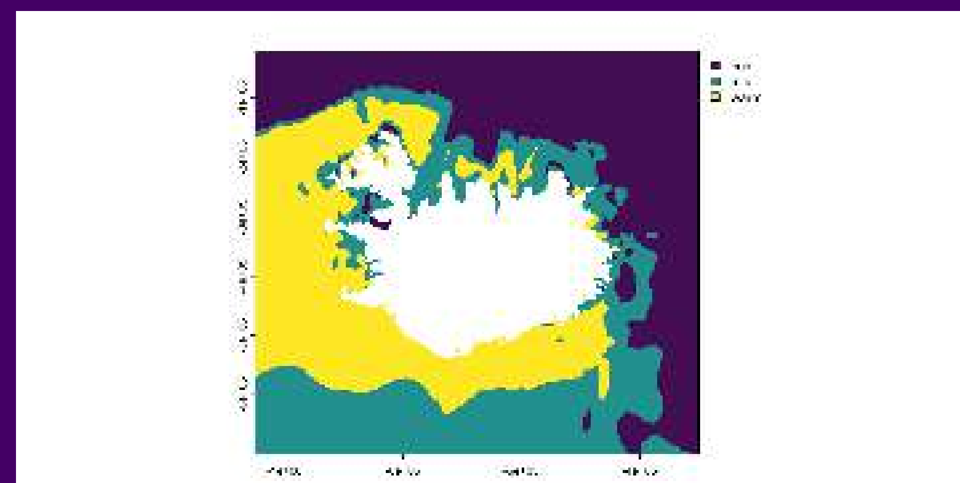


- **Regular**: constant cell size; axes aligned with Easting and Northing
- **Rotated**: constant cell size; axes not aligned with Easting and Northing
- **Sheared**: constant cell size; axes not parallel
- **Rectilinear**: cell size varies along a dimension
- **Curvilinear**: cell size and orientation dependent on the other dimension

Types of Raster Data

- **Continuous**: numeric data representing a measurement (e.g., elevation, precipitation)
- **Categorical**: integer data representing factors (e.g., land use, land cover)





What makes data (geo)spatial?

Location vs. Place

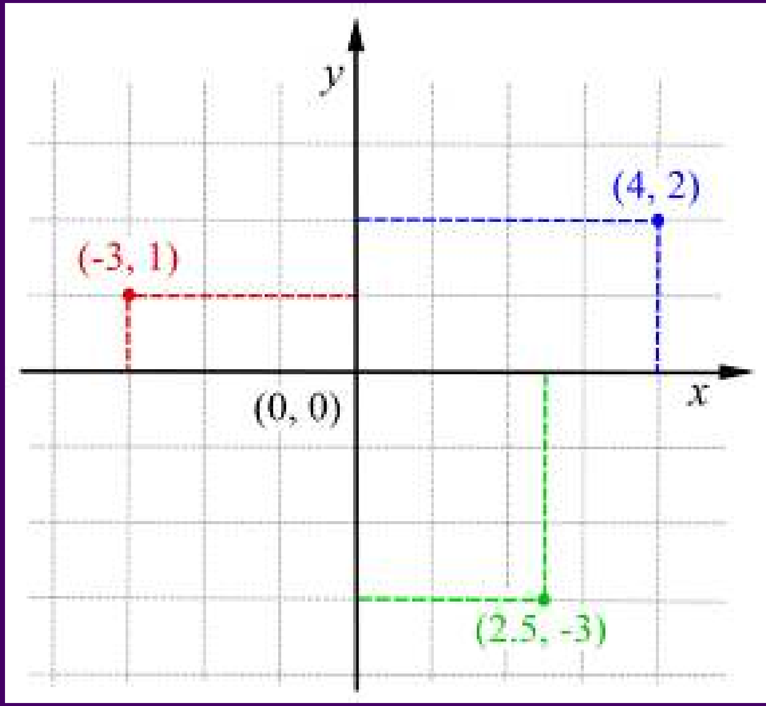
- **Place:** an area having unique **physical** and **human** characteristics **interconnected** with other places
- **Location:** the actual position on the earth's surface
- **Sense of Place:** the emotions someone attaches to an area based on experiences
- Place is *location plus meaning*

- **nominal: (potentially contested) place names**
- **absolute: the physical location on the earth's surface**

Describing Absolute Locations

- **Coordinates:** 2 or more measurements that specify location relative to a *reference system*

- Cartesian coordinate system
- *origin (O)* = the point at which both measurement systems intersect
- Adaptable to multiple dimensions (e.g. *z* for altitude)



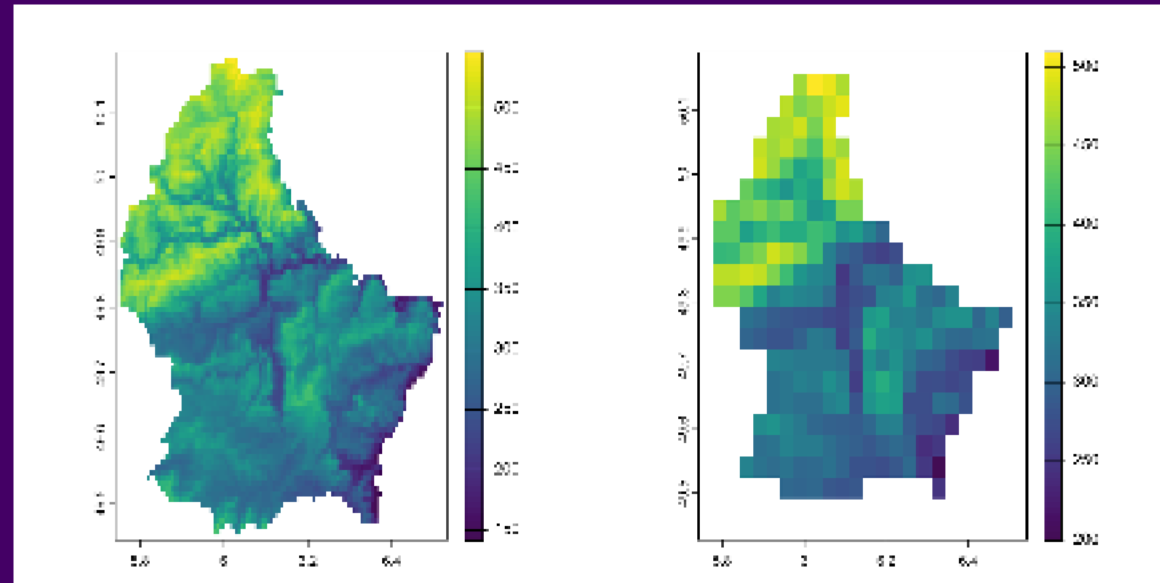
Cartesian Coordinate System

Describing location: extent

- How much of the world does the data cover?
- For rasters, these are the corners of the lattice
- For vectors, we call this the bounding box

Describing location: resolution

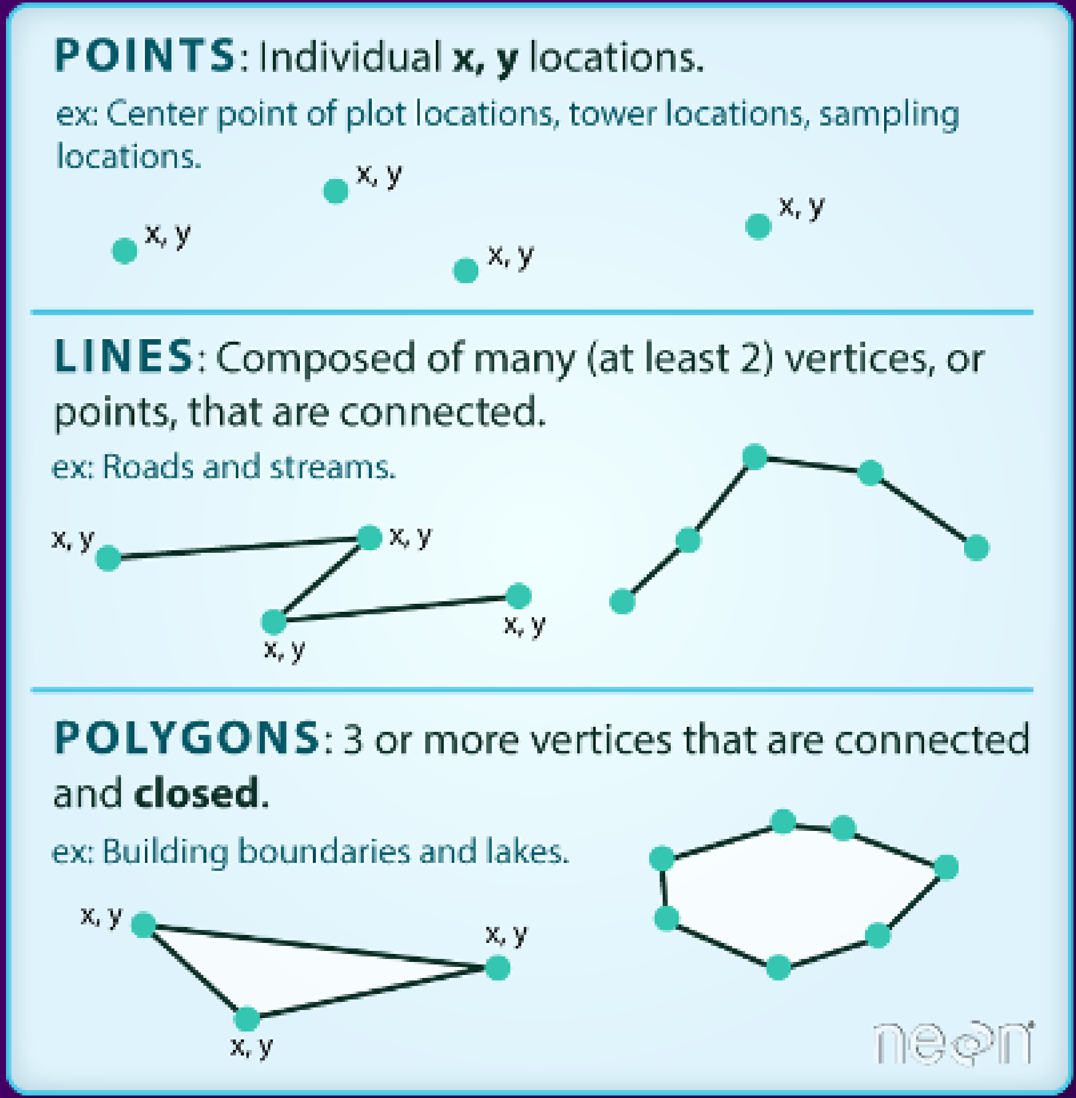
- **Resolution:** the accuracy that the location and shape of a map's features can be depicted
- **Minimum Mapping Unit:** The minimum size and dimensions that can be reliably represented at a given *map scale*.
- Map scale vs. scale of analysis



Geometries and Support

Geometries

- Vectors aggregate the locations of a feature into a geometry
- Most vector operations require simple, valid geometries



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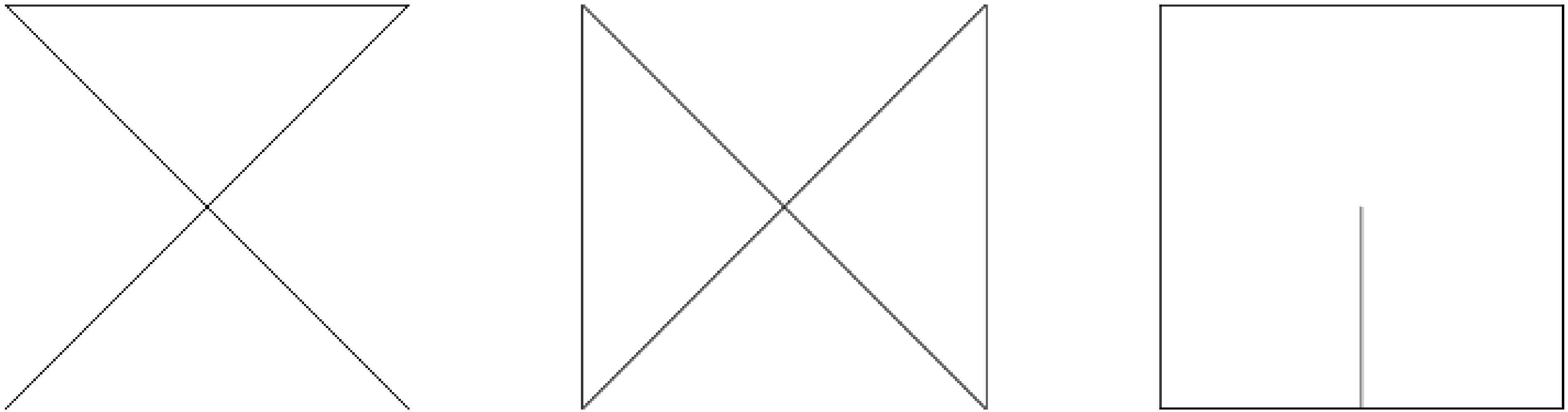
Valid Geometries

- A `linestring` is *simple* if it does not intersect
- Valid polygons
- Are closed (i.e., the last vertex equals the first)
- Have holes (inner rings) that inside the the exterior boundary
- Have holes that touch the exterior at no more than one vertex (they don't extend across a line) - For multipolygons, adjacent polygons touch only at points
- Do not repeat their own path



Empty Geometries

- Empty geometries arise when an operation produces `NULL` outcomes (like looking for the intersection between two non-intersecting polygons)
- `sf` allows empty geometries to make sure that information about the data type is retained
- Similar to a `data.frame` with no rows or a `list` with `NULL` values
- Most vector operations require simple, valid geometries

Support

Support is the area to which an attribute applies.

For vectors, the attribute-geometry-relationship can be:

- **constant** = applies to every point in the geometry (lines and polygons are just lots of points)
- **identity** = a value unique to a geometry
- **aggregate** = a single value that integrates data across the geometry

Support

Support is the area to which an attribute applies.

For rasters:

- **point** = attribute refers to the cell center
- **cell** = attribute refers to an area similar to the pixel

Take Homes

- Data models translate views of space *into* analytic tools
- Spatial data requires **location**
- We store location in geometries
- We assign attributes to location based on support